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# Adaptive Path Planning for Multi-UAV Systems in Dynamic 3D Environments: A Multi-Objective Framework

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**Abstract:** This study evaluates and compares the computational performance and practical applicability of advanced path planning algorithms for Unmanned Aerial Vehicles (UAVs) in dynamic and obstacle-rich environments. The Adaptive Multi-Objective Path Planning (AMOPP) framework is highlighted for its ability to balance multiple objectives, including path length, smoothness, collision avoidance, and real-time responsiveness. Through experimental analysis, AMOPP demonstrates superior performance, with a 15% reduction in path length compared to A*, achieving an average path length of 450 m. Its angular deviation of 8.0° ensures smoother trajectories than traditional methods like Genetic Algorithm and Particle Swarm Optimization (PSO). Moreover, AMOPP achieves a 0% collision rate across all simulations, surpassing heuristic-based methods like Cuckoo Search and Bee Colony Optimization, which exhibit higher collision rates. Real-time responsiveness is another key strength of AMOPP, with an average re-planning time of 0.75 s, significantly outperforming A* and RRT*. The computational complexities of each algorithm are analyzed, with AMOPP exhibiting a time complexity of $O(k \cdot n)$ and a space complexity of $O(n)$, ensuring scalability and efficiency for large-scale operations. The study also presents a comprehensive qualitative and quantitative comparison of 14 algorithms using 3D visualizations, highlighting their strengths, limitations, and suitable application scenarios. By integrating weighted optimization with penalty-based strategies and spline interpolation, AMOPP provides a robust solution for UAV path planning, particularly in scenarios requiring smooth navigation and adaptive re-planning. This work establishes AMOPP as a promising framework for real-time, efficient, and safe UAV operations in dynamic environments.

**Keywords:** multi UAV; multi path planning; path planning; multi-objective; A*

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## 1. Introduction

Unmanned Aerial Vehicles (UAVs) have become indispensable across numerous industries, including military surveillance, logistics, environmental monitoring, precision agriculture, and disaster response [1–3]. Their ability to operate autonomously in complex and hazardous environments, where direct human intervention is often challenging or impossible, has revolutionized traditional operational methods [4–6]. UAVs enable faster, more efficient, and cost-effective solutions for tasks ranging from data collection and cargo delivery to search and rescue missions [7]. However, as the scope of UAV applications continues to grow, so does the complexity of the operational challenges they face [8]. In particular, UAVs must navigate environments with unpredictable obstacles, limited navigable

space, and dynamic conditions, requiring advanced path planning capabilities to optimize mission performance [9]. The complexity increases further when multiple UAVs operate concurrently, as they must coordinate to avoid collisions while fulfilling their individual mission objectives [10]. Thus, the development of effective path planning algorithms that ensure optimal navigation in complex, real-world scenarios is crucial for maximizing the potential of UAV technology [11]. Traditional path planning techniques, such as A* and Dijkstra's algorithm, have provided foundational solutions for UAV navigation [12]. These algorithms use deterministic approaches that guarantee optimality in static, well-defined environments by finding the shortest path between a starting point and a destination [13]. However, their limitations become evident in real-world scenarios characterized by environmental dynamism and uncertainty. In such situations, UAVs require flexible, adaptive path planning to navigate safely [14]. Heuristic optimization methods like Particle Swarm Optimization (PSO) and Genetic Algorithms (GAs) have emerged as alternatives to address these limitations, providing a means to explore a broader solution space and effectively handle non-deterministic factors [15]. These methods allow for near-optimal solutions in complex environments but often at the cost of higher computational requirements, which can hinder real-time application [16]. Reinforcement Learning (RL) has further extended the capabilities of UAV path planning by enabling UAVs to learn optimal navigation policies through interaction with the environment, allowing real-time adaptation [17]. Despite these advances, most approaches still focus on single-objective optimization, such as minimizing travel distance or energy consumption, which falls short in addressing the multi-faceted and often conflicting objectives inherent in practical UAV missions.

The growing demand for UAV deployment in constrained environments such as urban areas, disaster zones, and mountainous regions has underscored the need for advanced multi-objective path planning [18]. In these challenging environments, UAVs must navigate through complex obstacles, avoid collisions, and ensure flight smoothness, all while achieving mission efficiency [19]. For instance, in urban settings, UAVs must avoid buildings and other infrastructure while adhering to regulations and minimizing noise and energy consumption. In disaster response scenarios, time is of the essence, and UAVs must reach their target locations quickly, often navigating through debris and unpredictable hazards [20–22]. The importance of real-time adaptability becomes even more critical in such time-sensitive applications, as delays or suboptimal paths can lead to mission failure or increased risks. Therefore, there is an urgent need for sophisticated path planning algorithms capable of optimizing multiple, sometimes conflicting, objectives, such as minimizing path length, improving flight smoothness, avoiding obstacles, and managing computational efficiency, while adapting to real-time changes in the environment [23].

Recent advancements in multi-objective optimization for UAV path planning have shown promise through the use of advanced techniques like Multi-Objective Particle Swarm Optimization (MOPSO) and Multi-Objective Genetic Algorithms (MOGAs) [24]. These methods enable UAVs to balance several criteria, such as energy efficiency, time minimization, and obstacle avoidance, by allowing for trade-offs between competing objectives. Unlike traditional single-objective optimization approaches, these algorithms provide a framework for more comprehensive mission planning that considers the full scope of operational requirements [25]. However, their performance in dynamic environments often suffers due to high computational demands, which make real-time implementation challenging. Reinforcement Learning (RL) approaches have also gained significant traction for their capacity to learn from the environment and adapt paths dynamically [26]. Despite their potential, RL-based methods are often limited by extensive training requirements, susceptibility to environmental instability, and scalability challenges in multi-UAV coordination [27]. Moreover, existing approaches frequently overlook the importance of path smoothness as a critical aspect of UAV path planning that directly affects vehicle stability, safety, and energy consumption. Abrupt changes in trajectory can lead to increased wear and tear on UAV components and higher energy usage, making smooth, gradual transitions essential for efficient operations [28]. Addressing these gaps requires the development of a

more robust and adaptive path planning approach that can effectively integrate multiple objectives while ensuring computational feasibility in real-time scenarios.

The primary research gap lies in achieving an efficient, real-time, multi-objective path planning solution for multiple UAVs operating in complex and dynamically changing environments. Most existing studies tend to simplify environmental conditions, either by reducing the number of obstacles or assuming static settings, which limits their applicability to real-world operations where adaptability is crucial [29,30]. Furthermore, computational inefficiencies in current models hinder their ability to be implemented in real-time, a key requirement for responsive and autonomous UAV operations. Path smoothness, a vital element for maintaining UAV stability, energy efficiency, and mission safety, is often inadequately addressed or entirely overlooked in these models. Smooth trajectories minimize abrupt changes in direction or altitude, reducing mechanical strain and optimizing energy use, which are essential for successful UAV missions in constrained environments [9,31,32]. To bridge these gaps, this research aims to develop an Advanced Multi-Objective Path Planner (AMOPP) that can balance multiple objectives effectively while maintaining adaptability and computational efficiency in real-time operations.

The proposed AMOPP framework is designed specifically for multi-UAV systems operating in Three-Dimensional (3D) environments characterized by high complexity and dynamism. The AMOPP focuses on optimizing key parameters, such as minimizing path length, enhancing path smoothness, avoiding collisions, and ensuring the overall feasibility of the planned trajectories. By integrating heuristic optimization methods with real-time adaptive strategies, the AMOPP provides UAVs with the flexibility required to adjust their paths dynamically in response to changing environmental conditions. This integration not only improves computational efficiency, making the system suitable for real-time application, but also addresses the practical challenges of UAV deployment in complex scenarios. The AMOPP will be evaluated through simulations that replicate real-world operational environments, including diverse obstacle fields and 3D terrains, demonstrating its adaptability, efficiency, and effectiveness in optimizing multiple objectives. The AMOPP framework is designed to optimize UAV missions such as urban delivery and disaster response, where objectives include minimizing travel time, ensuring safe navigation around obstacles, and optimizing energy usage for prolonged operations. The remainder of this journal article is structured as follows: The next section provides a review of the literature related to UAV path planning, focusing on recent developments in multi-objective optimization and multi-UAV coordination, and identifying the gaps this research seeks to address. The subsequent section details the methodology used in developing the AMOPP framework, including the mathematical formulation of the path planning problem and the optimization techniques employed. This section also elaborates on the computational mechanism, explaining how the proposed approach ensures real-time feasibility while balancing conflicting objectives. The experimental setup and results section then presents the performance of the AMOPP in simulated environments, highlighting its effectiveness in optimizing path length, smoothness, and collision avoidance, and its efficiency in real-time applications. It also provides insights into the practical implications of the results, comparing the proposed framework with existing methods and outlining its strengths and limitations. Finally, the conclusion summarizes the key findings of the research, emphasizing the contributions of the AMOPP to the field of UAV path planning, and suggests future directions for enhancing UAV coordination and adaptability in even more challenging operational contexts.

## 2. Literature Survey

Path planning for Unmanned Aerial Vehicles (UAVs) has been a critical area of research over the past two decades, as UAVs have evolved from military tools into essential components of civilian and commercial applications [33]. The complexity of UAV missions has grown in parallel with advances in autonomy and artificial intelligence, requiring the development of sophisticated algorithms capable of navigating complex, dynamic environ-

ments [34]. In particular, multi-objective optimization and multi-UAV coordination have become focal points in recent research, as UAV missions increasingly demand balancing multiple operational objectives and coordinating multiple vehicles within shared airspace. This section provides a review of the key developments in these areas, highlighting the progress made and identifying the gaps that remain.

### 2.1. Traditional Path Planning Methods and Their Limitations

Early UAV path planning research primarily focused on deterministic algorithms such as A*, Dijkstra's, and Rapidly-exploring Random Trees (RRTs), which were widely used for static and obstacle-free environments [35]. These methods aimed to find the shortest or least costly path between a starting point and a destination. The simplicity and determinism of these methods made them suitable for small-scale operations or controlled environments [36]. However, their inability to handle dynamic environments, adapt to real-time changes, or optimize multiple objectives limited their practical application in real-world UAV operations.

For instance, A* and Dijkstra's algorithm provide optimal solutions for shortest path problems but are computationally expensive and do not scale well in large environments, particularly when dynamic obstacles or multiple objectives, such as minimizing energy consumption and avoiding hazardous areas, need to be considered [37]. Similarly, RRT and its variants have been effective in generating feasible paths in cluttered spaces but tend to produce jerky paths that may not be suitable for smooth UAV flight [38,39]. These traditional methods often assume static environments and single-objective optimization, which restricts their adaptability and efficiency in complex, real-time scenarios involving multiple UAVs.

### 2.2. Emergence of Heuristic Optimization Techniques

Recognizing the limitations of traditional path planning methods, researchers have turned to heuristic optimization algorithms that allow for more flexibility in handling non-deterministic and complex environments [40]. Heuristic methods, such as PSO, GA, ACO, and SA, have proven effective in exploring large search spaces and providing near-optimal solutions to multi-objective problems [41]. These methods excel in balancing multiple constraints, such as minimizing travel time, energy consumption, and ensuring collision avoidance.

PSO, for example, mimics the social behavior of birds flocking to find optimal paths, allowing UAVs to collectively find efficient routes while considering constraints like obstacle avoidance and energy efficiency [42]. GAs, inspired by the process of natural selection, have been extensively used to generate diverse path solutions, allowing UAVs to evolve paths through iterative crossover and mutation operations [43]. ACO has been another popular approach, especially for multi-UAV coordination, as it mimics the way ants find the shortest paths between their nests and food sources [44]. These methods provide more adaptive solutions than traditional path planning approaches, especially in environments where dynamic changes and multiple objectives must be considered simultaneously.

However, while heuristic optimization techniques have enabled UAVs to address more complex and dynamic path planning scenarios, they still face several challenges. One of the main limitations is computational overhead, which can hinder real-time application, particularly for large-scale operations involving multiple UAVs. In addition, heuristic methods often converge on local optima rather than global optima, which can result in sub-optimal performance in certain environments. As a result, while heuristic optimization has been a significant advancement over traditional methods, there is still a need for more robust solutions capable of real-time optimization and better handling of dynamic, multi-UAV scenarios.

### 2.3. Multi-Objective Optimization in UAV Path Planning

In recent years, research has increasingly focused on Multi-Objective Optimization (MOO) for UAV path planning, as real-world missions typically involve balancing several conflicting objectives. For instance, UAVs must minimize both flight time and energy consumption while avoiding obstacles and ensuring path smoothness for stable flight. Multi-objective optimization techniques, such as MOPSO and MOGA, have been employed to address these challenges by allowing UAVs to trade off between different objectives [45].

MOPSO extends the basic PSO algorithm by introducing Pareto optimality, which enables UAVs to explore a range of trade-offs between objectives [46]. MOPSO has been particularly effective in handling problems with two or more conflicting objectives, such as minimizing path length while avoiding high-risk areas or minimizing energy use while maintaining a certain altitude [47]. Similarly, MOGA applies evolutionary principles to generate a diverse set of solutions, each representing a different trade-off between objectives. By maintaining a Pareto front, MOGA ensures that solutions are spread across the trade-off space, providing UAVs with multiple viable paths depending on the mission's priorities [48].

Despite their strengths, multi-objective optimization approaches still face significant challenges, particularly in real-time operations. The computational complexity of maintaining and updating a Pareto front can slow down decision-making processes, which is especially problematic when UAVs need to adapt to rapidly changing environments [49]. Additionally, most existing MOO algorithms focus primarily on optimizing objectives such as distance and energy consumption, often overlooking critical factors like path smoothness, which is vital for ensuring stable flight and minimizing mechanical strain on UAVs [50,51]. This gap highlights the need for more comprehensive MOO frameworks that can address a wider range of objectives while maintaining real-time applicability.

### 2.4. Recent Advances in DRL-Aided UAV Navigation

Deep Reinforcement Learning (DRL) has recently gained attention for its ability to tackle complex, dynamic UAV navigation tasks by learning optimal policies from experience. Notable advancements include quantum-inspired experience replay techniques, which enhance learning efficiency and convergence in challenging environments [52]. These methods allow UAVs to adapt to real-time changes more effectively than traditional or heuristic approaches. Despite their potential, DRL-based methods require significant computational resources and extensive training, which remain barriers to widespread adoption in real-world scenarios.

### 2.5. Contributions of This Work

To address the gaps identified in the existing research, this work makes the following contributions:

- Proposes an Adaptive Multi-Objective Path Planning (AMOPP) framework, incorporating weighted optimization and penalty-based strategies.
- Introduces spline interpolation for enhanced path smoothness and stability.
- Evaluates computational efficiency and scalability in dynamic, multi-UAV scenarios.
- Demonstrates superior performance compared to 14 state-of-the-art algorithms through rigorous experimental analysis.

### 2.6. Gaps and Future Directions

Despite the significant advancements in UAV path planning, several gaps remain. One of the most prominent is the lack of comprehensive multi-objective optimization frameworks that can handle both traditional metrics (such as path length and energy consumption) and more nuanced objectives (such as path smoothness and stability) [53]. Existing multi-objective optimization techniques, while powerful, often struggle with computational efficiency, making them difficult to apply in real-time operations, especially in dynamic environments involving multiple UAVs [54].

Additionally, while heuristic and AI-based methods have significantly advanced the field, they are often computationally expensive and prone to converging on local optima [55–58]. This highlights the need for novel approaches that combine the strengths of heuristic optimization with real-time adaptability to ensure UAVs can respond quickly and effectively to environmental changes. Lastly, more research is needed to explore the integration of path smoothness and UAV stability into optimization frameworks, as these factors are essential for long-term UAV operation and mission success.

## 3. Proposed Method

### *3.1. Problem Formulation*

The path planning problem for each UAV is framed as a continuous optimization problem, where the UAV's trajectory is represented as a continuous curve in three-dimensional space. Let $\mathbf{T}(t) = (x(t), y(t), z(t)) \in \mathbb{R}^3$ represent the position of a UAV at time $t$, where $x(t), y(t), z(t)$ are the Cartesian coordinates. The primary challenge is to find an optimal trajectory that satisfies several objectives while obeying the physical constraints of the UAV.

One of the fundamental objectives is to minimize the total path length. This can be expressed mathematically as an integral over the trajectory from the starting point to the final destination. The path length, $L$, is given by the integral of the instantaneous velocity magnitude over time, expressed as in Equation (1).

$$L = \int_0^{T_f} \sqrt{\left(\frac{dx(t)}{dt}\right)^2 + \left(\frac{dy(t)}{dt}\right)^2 + \left(\frac{dz(t)}{dt}\right)^2}\, dt \tag{1}$$

This formulation ensures that the UAV follows the shortest feasible path between the source and target locations, where $T_f$ denotes the final time of the UAV's trajectory. This objective ensures that the UAV minimizes its total travel distance, which directly impacts energy consumption and operational efficiency. In addition to minimizing path length, it is crucial to ensure the smoothness of the UAV's path. Path smoothness is essential for maintaining stable flight, reducing mechanical wear on the UAV, and optimizing energy consumption. Sharp turns or abrupt changes in direction can lead to increased energy usage and mechanical stress. The smoothness of the trajectory is quantified by minimizing the rate of change of the UAV's angular orientation. Let the UAV's velocity vector be defined as $\mathbf{v}(t) = (\dot{x}(t), \dot{y}(t), \dot{z}(t))$, where $\dot{x}(t) = \frac{dx(t)}{dt}$, and similarly for the other components. The smoothness criterion is expressed by minimizing the rate of change of the angles of the trajectory, which can be formulated as in Equation (2).

$$f_{\text{smooth}}(\mathbf{T}) = \int_0^{T_f} \left| \frac{d}{dt} \left( \arctan\left( \frac{\sqrt{\dot{x}(t)^2 + \dot{y}(t)^2}}{\dot{z}(t)} \right) \right) \right| dt \tag{2}$$

This formulation captures the changes in the UAV's direction of motion, where minimizing this term leads to smoother flight paths with fewer abrupt directional changes. In addition, this expression captures the changes in the UAV's direction of motion, where minimizing this term leads to smoother flight paths with fewer abrupt directional changes. Another critical aspect of the AMOPP framework is ensuring collision avoidance. The operational environment often contains obstacles that the UAVs must avoid while navigating. These obstacles are modeled as spheres with known centers and radii. Let $\mathbf{O}_k = (O_x^k, O_y^k, O_z^k)$ denote the position of the $k$-th obstacle and $r_k$ its radius. In practical terms, the obstacle is generated from sensor and obstacle modeling. Then, in order to avoid the collision, a penalty function, $P_k(\mathbf{T})$, is introduced to enforce collision avoidance. This function assigns a penalty if the UAV's trajectory comes within the radius of an obstacle, as presented in Equation (3).

$$P_k(\mathbf{T}) = \sum_{t=0}^{T_f} H(r_k - \|\mathbf{T}(t) - \mathbf{O}_k\|) \tag{3}$$

where $H(x)$ is the Heaviside step function, which assigns an infinite penalty if the distance between the UAV and the obstacle is less than $r_k$. This ensures that any trajectory violating the collision constraint is infeasible, thereby preventing UAVs from entering restricted or hazardous zones. In addition, feasibility is defined in terms of the UAV's physical motion constraints, including velocity and acceleration limits. The maximum velocity, $v_{\max}$, and acceleration, $a_{\max}$, are enforced as hard constraints on the UAV's motion. Energy efficiency, another important consideration, is modeled as a function of the UAV's velocity. The power consumption, $P(t)$, at time $t$ is given by (4).

$$P(t) = P_0 + P_v \|\mathbf{v}(t)\|^2 \tag{4}$$

where $P_0$ is the baseline power consumption for operating the UAV and $P_v$ is a proportionality constant reflecting the additional power required for motion. Minimizing the total energy consumed over the flight path is crucial for extending the UAV's operational duration, particularly in energy-constrained scenarios. The path planning problem, therefore, becomes a multi-objective optimization problem that seeks to simultaneously minimize path length, maximize smoothness, avoid collisions, and ensure energy efficiency. The formal mathematical expression of this multi-objective problem is presented in Equation (5).

$$\min_{\mathbf{T}(t)} \Big( f_{\text{length}}(\mathbf{T}), f_{\text{smooth}}(\mathbf{T}), f_{\text{collision}}(\mathbf{T}), f_{\text{energy}}(\mathbf{T}) \Big) \tag{5}$$

This optimization problem is subject to constraints on the UAV's dynamics and boundary conditions, where $\mathbf{T}(0) = \mathbf{T}_{\text{start}}$ and $\mathbf{T}(T_f) = \mathbf{T}_{\text{goal}}$. These constraints ensure that the UAV starts from a designated source location and reaches its target while maintaining the desired motion characteristics. To solve this complex multi-objective problem, the AMOPP framework employs a weighted aggregation approach, where each objective is assigned a weight reflecting its relative importance. The weighted sum of the objective functions is minimized, as presented in Equation (6).

$$\begin{aligned} F(\mathbf{T}) = {} & \lambda_1 f_{\text{length}}(\mathbf{T}) + \lambda_2 f_{\text{smooth}}(\mathbf{T}) \\ & + \lambda_3 f_{\text{collision}}(\mathbf{T}) + \lambda_4 f_{\text{energy}}(\mathbf{T}) \end{aligned} \tag{6}$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \in [0, 1]$ are weight coefficients that sum to one and reflect the mission priorities. By adjusting these weights, the AMOPP can prioritize different objectives based on specific mission requirements.

### *3.2. AMOPP Algorithm*

The AMOPP algorithm optimizes UAV trajectories through a continuous, iterative refinement process. The objective is to balance multiple conflicting goals, such as minimizing path length, improving smoothness, ensuring collision avoidance, and maximizing energy efficiency. The optimization framework is based on a cost function that integrates these objectives into a single scalar value to be minimized. In the initial phase, the UAV's trajectory is represented by a set of discrete points in a 3D space. Then, the trajectory is discretized into $n$ segments, with the points $\mathbf{p}_i = (x_i, y_i, z_i)$ representing the UAV's position at the $i$-th time step, where $i = 1, 2, \ldots, n-1$.

The algorithm starts by constructing a linear trajectory from the start point, $\mathbf{s} = (x_s, y_s, z_s)$, to the goal point, $\mathbf{g} = (x_g, y_g, z_g)$. The initial path, $\mathbf{P}_{\text{init}}$, is represented by the linearly interpolated points between $\mathbf{s}$ and $\mathbf{g}$, ensuring a feasible starting solution for the optimization process. This trajectory is then refined iteratively by perturbing the intermediate points in the path and evaluating the effect of these perturbations on the overall cost. At each iteration, the intermediate points, $\mathbf{p}_i$, for $i = 1, 2, \ldots, n-1$, are perturbed by adding a random vector, $\mathbf{r} \sim U(-1, 1)^3$, where $U(-1, 1)$ is a uniform distribution over the interval $[-1, 1]$. The updated point is given by Equation (7).

$$\mathbf{p}_i' = \mathbf{p}_i + \mathbf{r} \tag{7}$$

This new candidate point is constrained within the predefined environment boundaries, ensuring the UAV remains within the operational space, as presented in Equation (8).

$$\mathbf{p}_i' \in [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}] \tag{8}$$

If the new candidate point lies within a no-fly zone or violates obstacle constraints, it is rejected, and the original point, $\mathbf{p}_i$, is retained for that iteration. Otherwise, the new path is evaluated for its cost. The cost function, $C(\mathbf{P})$, incorporates multiple objectives: path length, smoothness, collision avoidance, and energy efficiency. The path length, $L(\mathbf{P})$, is defined as the integral of the instantaneous velocity magnitude over the trajectory, as presented in (1). In addition, path smoothness is another critical objective, quantified by the rate of change of the UAV's angular orientation. The smoothness criterion is defined as the minimization of the angular rate of change, as presented in (2). This ensures that the UAV's trajectory avoids sharp turns and abrupt directional changes, leading to smoother flight paths and reduced mechanical stress. Collision avoidance is enforced by introducing a penalty function, $P_k(\mathbf{T})$, which penalizes trajectories that pass within the radius of any obstacle. Obstacles are modeled as spheres with known centers, $\mathbf{O}_k = (O_x^k, O_y^k, O_z^k)$, and radii, $r_k$. The penalty function is given in (3) and the power consumption, $P(t)$, at time $t$ is expressed as (4). Therefore, the overall cost function is then defined as Equation (9).

$$C(\mathbf{P}) = L(\mathbf{P})(1 + 100 \cdot V(\mathbf{P})) + E(\mathbf{P}) \tag{9}$$

where $V(\mathbf{P})$ is the total violation of constraints (such as collision avoidance) and $E(\mathbf{P})$ is the total energy cost over the trajectory. The total violation, $V(\mathbf{P})$, is computed by summing the penalties for all obstacles along the path. The energy cost, $E(\mathbf{P})$, is proportional to the path length, ensuring that shorter paths consume less energy. After evaluating the cost of the updated trajectory, the algorithm compares the new cost with the cost of the current best path. If the new cost is lower, the new path is accepted as the best path; otherwise, the previous path is retained. The AMOPP algorithm also updates the positions of dynamic obstacles and moving targets at each iteration. The position of each dynamic obstacle is perturbed according to a uniform distribution, as presented in Equation (10).

$$\mathbf{O}_k' = \mathbf{O}_k + \mathbf{v}_k, \quad \mathbf{v}_k \sim U(-v_{\max}, v_{\max})^3 \tag{10}$$

where $v_{\max}$ is the maximum velocity of the obstacle. This ensures that the algorithm can adapt to changes in the environment, providing real-time path planning capabilities. The optimization process is repeated for a fixed number of iterations or until convergence. The details of the algorithms implemented are presented in the pseudocode, as described in Algorithms 1–5.

In addition, to address the issue of communication overhead, the proposed Adaptive Multi-Objective Path Planning (AMOPP) framework inherently operates in a decentralized manner, minimizing the need for inter-agent communication. Each UAV calculates its trajectory independently based on the global constraints defined in the environment model. This decentralized optimization ensures that the computational load and communication requirements are distributed across all agents, rather than centralized in a single coordinating entity. Mathematically, each UAV, $i$, optimizes its path by minimizing the cost function, $C_i$, defined as a function of trajectory smoothness, $L$, collision violations, $V$, and other objectives, such that $C_i = L_i \cdot (1 + \lambda V_i)$, where $\lambda$ is a penalty weight. Since each UAV independently computes $C_i$ using only its local trajectory information and obstacle data, the need for real-time communication is reduced to critical updates, such as sharing positional data during potential trajectory conflicts. The communication protocol is represented as $\mathcal{M}(p_i, p_j)$, where $p_i$ and $p_j$ are the positions of UAVs $i$ and $j$, ensuring that only essential state data is exchanged when inter-UAV distances fall below a predefined threshold.

**Algorithm 1:** AMOPP Main Path Optimization

**Input:** start: Starting point $(x_s, y_s, z_s)$,
goal: Goal point $(x_g, y_g, z_g)$,
model: Environment model,
num_segments: Number of path segments,
iterations: Maximum number of optimization iterations,
bounds: Operational space boundaries $[x_{\min}, x_{\max}, y_{\min}, y_{\max}, z_{\min}, z_{\max}]$
**Output:** optimal_path: Optimized UAV path

```
Initialize initial_path as a linear interpolation between start and goal with
  num_segments segments;
best_path ← initial_path;
best_cost ← CalculateCost(best_path, start, goal, model);
for iter from 1 to iterations do
    UpdateDynamicObstacles(model);
    UpdateMovingTargets(model);
    for i from 1 to num_segments - 1 do
        candidate_point ← best_path[i] + RandomPerturbation (-1, 1);
        candidate_point ← Clip(candidate_point, bounds);
        if not InNoFlyZone(candidate_point, model) then
            new_path ← best_path;
            new_path[i] ← candidate_point;
            new_cost ← CalculateCost(new_path, start, goal, model);
            if new_cost < best_cost then
                best_path ← new_path;
                best_cost ← new_cost;
optimal_path ← best_path;
return optimal_path
```

**Algorithm 2:** Cost Calculation for AMOPP Path

**Input:** path: Current UAV path,
start: Start point,
goal: Goal point,
model: Environment model containing obstacles and boundaries
**Output:** cost: Calculated cost of the path

```
length ← ComputePathLength(path);
violation ← ComputeCollisionViolation(path, model);
energy_cost ← ComputeEnergyCost(path);
cost ← length × (1 + 100 × violation) + energy_cost;
return cost
```

**Algorithm 3:** Dynamic Obstacle Update

**Input:** model: Environment model with obstacles
**Output:** Updated obstacle positions in the model

```
for each obstacle in model do
    obstacle.position ← obstacle.position + RandomPerturbation (-max_speed,
      max_speed);
    obstacle.position ← Clip(obstacle.position, bounds);
```

**Algorithm 4:** Moving Target Update

**Input:** model: Environment model with moving targets
**Output:** Updated target positions in the model
**for** *each target in model* **do**
    target.position ← target.position + `RandomPerturbation` (-target_speed, target_speed);
    target.position ← `Clip`(*target.position, bounds*);

**Algorithm 5:** Check for No-Fly Zone Violation

**Input:** point: The UAV's current position,
model: Environment model with no-fly zones
**Output:** Boolean value indicating whether the point is in a no-fly zone
no_fly_zone_center ← [25, 25, 25];
no_fly_zone_radius ← 5;
distance ← EuclideanDistance(point, no_fly_zone_center);
**return** *distance* < *no_fly_zone_radius*

For handling sparse or delayed rewards, the AMOPP framework employs a well-designed cost function that directly incorporates feedback during the optimization process. The cost function provides immediate penalties for collisions and rewards for smooth trajectories, effectively shaping the optimization landscape without relying on terminal rewards. The path length, $L$, is calculated using the spline-interpolated trajectory, where the interpolated points $(x(t), y(t), z(t))$ are derived from the cubic spline representation. The path smoothness is further evaluated through the derivatives of the spline, which measure abrupt changes in direction, represented as $\int_0^1 \| \frac{d^2\mathbf{r}(t)}{dt^2} \| dt$, where $\mathbf{r}(t) = [x(t), y(t), z(t)]$. This continuous evaluation ensures that the UAV is incentivized to follow a smoother path at each iteration. The cost function also incorporates collision penalties, $V$, calculated as $V = \int_0^1 \max(1 - \frac{d(t)}{r_{\text{obs}}}, 0) dt$, where $d(t)$ is the distance between the UAV and the nearest obstacle and $r_{\text{obs}}$ is the obstacle radius. By integrating $V$ into the cost function, the framework effectively penalizes trajectories that come close to obstacles, providing real-time feedback on infeasible paths. These mechanisms address the sparse reward issue by ensuring that feedback is provided at every optimization step, avoiding the need to wait for terminal states to evaluate performance. Furthermore, the AMOPP framework iteratively refines each UAV's trajectory by sampling candidate points along the path and evaluating the resulting cost. Suppose $P = \{p_0, p_1, \ldots, p_n\}$ represents the trajectory points, where $p_0$ and $p_n$ are the source and destination, respectively. For each intermediate point, $p_i$, a candidate $p_i'$ is sampled within the bounds of the environment, $\mathcal{B}$, such that $p_i' \in \mathcal{B}$. The updated path, $P'$, is then evaluated, and if $C(P') < C(P)$ the candidate path is accepted. This iterative refinement continues until convergence, defined as $|C(P^{(k+1)}) - C(P^{(k)})| < \epsilon$, where $\epsilon$ is a small threshold. The least squares method is employed within the AMOPP framework to minimize the integral of acceleration squared over the UAV trajectory, a measure often associated with reducing jerk. This approach enhances path smoothness, critical for stable UAV operations and energy efficiency. For all simulations, the UAVs are assumed to start from rest $v_0 = 0$, a standard assumption for missions initiated from fixed takeoff points.

## 4. Experiment Setup

The experimental evaluation of the Advanced Multi-Objective Path Planner (AMOPP) was conducted using a series of simulations designed to assess its performance in optimizing key metrics, including path length, smoothness, and collision avoidance, as well as its efficiency in real-time applications. This section describes the simulation environment, the evaluation metrics used, and the results obtained from the experiments, demonstrating the effectiveness of the AMOPP framework in addressing the complex challenges of multi-UAV path planning.

### *4.1. Initial Setup*

To evaluate the AMOPP framework, a simulated three-dimensional environment was developed, consisting of dynamic and static obstacles to reflect the complexities encountered in real-world UAV missions. The environment spanned a region of 500 m × 500 m × 200 m, populated with spherical obstacles of varying radii ranging from 5 m to 15 m, randomly distributed throughout the space. These obstacles represented potential hazards such as buildings, trees, and other UAVs, with both static and dynamic properties to simulate real-time changes. Several UAVs (3–5) were deployed simultaneously in this environment, each assigned unique start and target locations to prevent overlapping paths. Each UAV operated under constraints on velocity and acceleration, consistent with commercial UAV capabilities, with a maximum velocity of 10 m/s and a maximum acceleration of 2 $m/s^2$. These constraints ensured realistic simulation conditions reflective of small urban UAV missions.

Changing flight conditions were incorporated within the experiment to emulate real-world operational challenges. Natural variations such as sudden wind speed changes, turbulence, and rainfall intensity were modeled to test UAV adaptability under unpredictable environmental conditions. Additionally, the terrain characteristics, including uneven altitudes and rugged surfaces, required the continuous recalibration of UAV trajectories to optimize energy consumption and maintain flight stability. Artificial challenges such as moving obstacles, shifting debris, and evolving no-fly zones were dynamically introduced into the environment to simulate the constraints encountered in urban and disaster-response missions. These conditions collectively influenced the optimization objectives, forcing the AMOPP framework to adapt in real-time by recalibrating flight paths to ensure safety, energy efficiency, and mission success.

The AMOPP framework was implemented using Python, and the simulations were executed on a high-performance system equipped with an Intel Core i9 processor and 32 GB of RAM to facilitate real-time evaluations. To validate distributed decision-making, each UAV performed its path planning on an independent simulated computing unit. This setup ensured that the optimization process operated in a decentralized manner, closely resembling real-world UAV deployments. The combination of a dynamically changing environment, adaptive optimization algorithms, and realistic UAV constraints allowed for a comprehensive evaluation of the AMOPP framework under varying flight conditions. The adaptive mechanisms integrated into the framework enabled it to recalibrate trajectories continuously, demonstrating its robustness and effectiveness in handling the complexities of real-world UAV missions.

### *4.2. Evaluation Metrics*

To evaluate the performance of the AMOPP framework, several key metrics were used to capture different aspects of the UAV's path planning. The first metric is the Path Length, which represents the total distance traveled by each UAV from its starting point to the target location. Mathematically, the path length, $L$, can be expressed as the integral of the UAV's velocity over time, as presented in Equation (1).

This serves as an indicator of the efficiency of the planned trajectory, where shorter paths suggest more optimal solutions in terms of travel time and energy consumption. In addition to path length, the Path Smoothness was measured by calculating the cumulative changes in direction along the path. The smoothness metric, $S$, can be defined as the sum of the changes in the heading angle, $\theta$, over the trajectory, calculated at each time step, as presented in Equation (11).

$$S = \sum_{i=1}^{n-1} |\theta_{i+1} - \theta_i| \tag{11}$$

Smoother and more stable paths are characterized by lower values of $S$, indicating fewer sharp turns or sudden movements. Furthermore, Collision Avoidance was another critical metric. This metric recorded the number of instances where a UAV's trajectory violated the obstacle clearance rule, meaning it came within the radius, $r_o$, of any ob-

stacle. The condition for a collision violation at time $t$ can be expressed as presented in Equation (12).

$$\|\mathbf{T}(t) - \mathbf{O}(t)\| < r_o \tag{12}$$

where $\mathbf{T}(t)$ represents the UAV's position at time $t$ and $\mathbf{O}(t)$ represents the position of an obstacle at time $t$. The goal is to achieve a zero value for this metric, indicating successful avoidance of all potential collisions. Lastly, Real-Time Feasibility was measured by assessing the amount of time, $\Delta t_{\text{plan}}$, each UAV took to compute a new trajectory in response to environmental changes. Real-time feasibility is achieved if Equation (13) is satisfied.

$$\Delta t_{\text{plan}} < 1\,\text{s} \tag{13}$$

A response time of under 1 s ensures the UAVs can adapt quickly to dynamic conditions. These metrics together provide a comprehensive assessment of the UAV's performance in terms of efficiency, stability, safety, and responsiveness.

## 5. Results and Discussions

### *5.1. Qualitative Analysis of Path Planning Results Through 3D Visualization*

In the first set of algorithms, as presented in Figure 1, DE produces paths that are characterized by unpredictable changes in direction, reflecting its exploratory nature. While DE successfully navigates around obstacles, the paths tend to be longer and less smooth compared to other algorithms. Similarly, the Firefly algorithm generates highly erratic paths with substantial loops and deviations. This chaotic behavior is indicative of the algorithm's reliance on randomness, which results in lower efficiency and suboptimal path smoothness. The GA, in contrast, produces relatively structured paths that balance exploration with a degree of refinement, resulting in moderate smoothness and shorter path lengths. This performance makes GA a suitable candidate for obstacle-rich environments where efficient navigation is required. Harmony Search, like Firefly, shows significant deviations in its path planning, often resulting in convoluted routes with unnecessary loops. The PSO algorithm, however, stands out for its ability to produce smooth and direct paths with minimal deviations, making it highly effective for obstacle-rich environments. PSO's balance between exploration and exploitation is evident in its efficient navigation, achieving paths that are both short and smooth. Finally, SA exhibits variability, with some segments of its paths following direct routes while others involve excessive exploration. Although SA effectively avoids obstacles, its paths are generally longer and less smooth, which reduces its efficiency compared to PSO or GA.

In the second set of algorithms, as presented in Figure 2, the Whale Optimization Technique achieves smooth, direct paths that navigate obstacles effectively, similar to PSO. Its paths reflect an efficient balance of exploration and exploitation, yielding a favorable trade-off between path length and obstacle avoidance. Cuckoo Search, however, exhibits convoluted paths with numerous loops, suggesting an emphasis on exploration over precision. Consequently, it struggles with path smoothness and length, making it less effective for environments requiring efficient obstacle navigation. The Bee Colony algorithm, akin to Cuckoo Search, generates highly convoluted paths that, while collision-free, are inefficient due to random deviations and loops. This emphasizes the algorithm's exploratory nature, resulting in lower suitability for structured path planning. ACO displays slightly smoother paths than Bee Colony, with fewer loops and chaotic deviations, though the paths remain longer than optimal. ACO's obstacle navigation is more refined, though it still falls short of PSO or Whale Optimization in terms of path smoothness and efficiency. MOGA performs similarly to GA but with an added advantage of handling multiple objectives, resulting in structured, direct paths that effectively avoid obstacles. MOGA demonstrates high efficiency, balancing path length, smoothness, and obstacle avoidance. MOPSO further enhances performance, producing some of the best paths in terms of smoothness, directness, and efficient obstacle navigation. Its paths are shorter and smoother than those generated by other multi-objective algorithms, reflecting its strong suitability for complex environments where multiple objectives must be balanced.

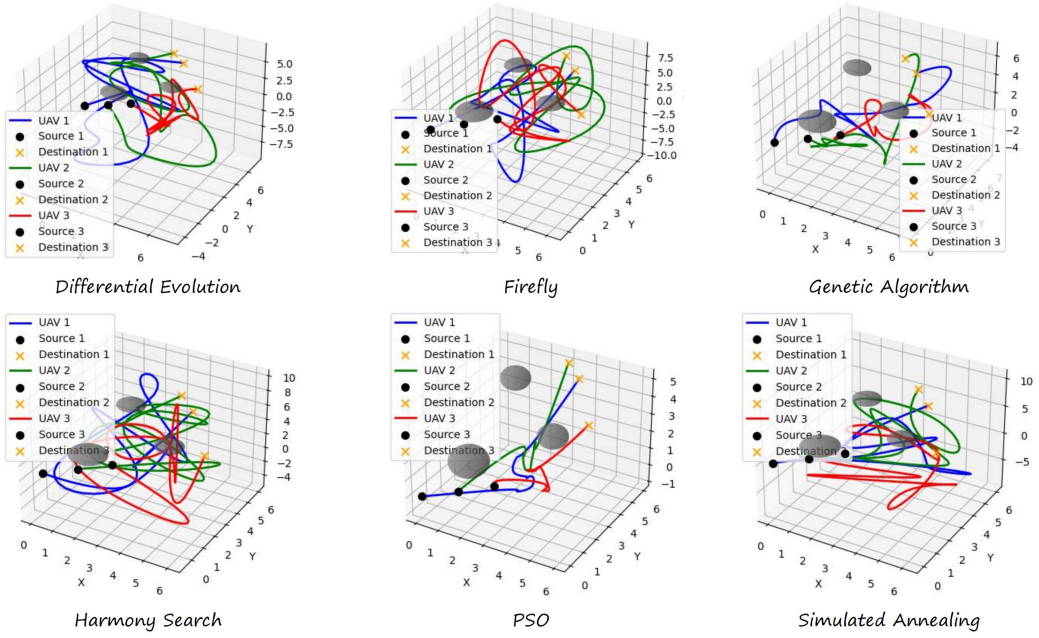



**Figure 1.** 3D path planning results for various metaheuristic and swarm intelligence algorithms. Each plot illustrates the paths planned by three UAVs (UAV 1, UAV 2, UAV 3) using different algorithms: Differential Evolution, Firefly, Genetic Algorithm, Harmony Search, Particle Swarm Optimization, and Simulated Annealing. Obstacles are represented by gray spheres, and each UAV's path is color-coded for clear differentiation. The algorithms demonstrate varying path smoothness, obstacle avoidance efficiency, and path optimization capabilities.

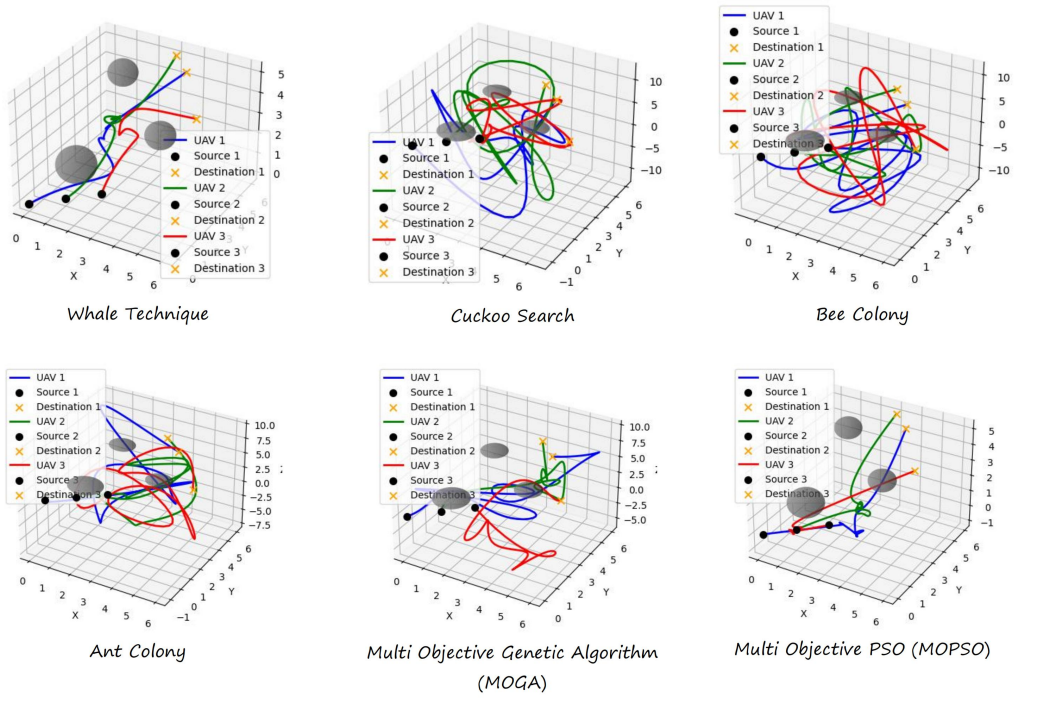



**Figure 2.** 3D path planning results for swarm intelligence and multi-objective optimization algorithms. This set illustrates the paths generated by various algorithms: Whale Optimization, Cuckoo Search, Bee Colony, Ant Colony Optimization, MOGA, and MOPSO. Each plot shows the paths planned by three UAVs (UAV 1, UAV 2, UAV 3) while navigating around obstacles represented by gray spheres. The paths demonstrate differences in smoothness, efficiency, and obstacle avoidance strategies depending on the algorithm used.

In Figure 3, the performance of deterministic approaches such as A* and AMOPP is highlighted. A* produces highly direct paths, achieving minimal deviation and exceptional obstacle avoidance. Its paths are among the shortest, underscoring A*'s optimality in structured environments. This efficiency makes A* an excellent choice for environments where the position of obstacles is known and direct navigation is prioritized. AMOPP, while producing slightly longer paths than A*, emphasizes smoothness, providing gentle curvatures around obstacles that may be beneficial in applications requiring gradual turns. AMOPP's balance between path length and smoothness makes it effective for scenarios where smooth navigation is prioritized over strict path length minimization. The comparison highlights the fact that algorithms like A*, MOPSO, and PSO perform exceptionally well, producing smooth, short, and obstacle-avoiding paths with high efficiency. AMOPP and MOGA offer a more balanced solution, providing paths that are both smooth and efficient, making them suitable for applications that require gradual and flexible navigation. In contrast, algorithms such as Firefly, Cuckoo Search, and Harmony Search exhibit a chaotic approach, with convoluted paths, which reduces their suitability for structured path planning tasks. The table encapsulates this performance differentiation, revealing the strengths of each algorithm in terms of path length, smoothness, obstacle avoidance, and overall efficiency, providing a comprehensive understanding of their capabilities and limitations. The summary of each algorithm performance is presented in Table 1.

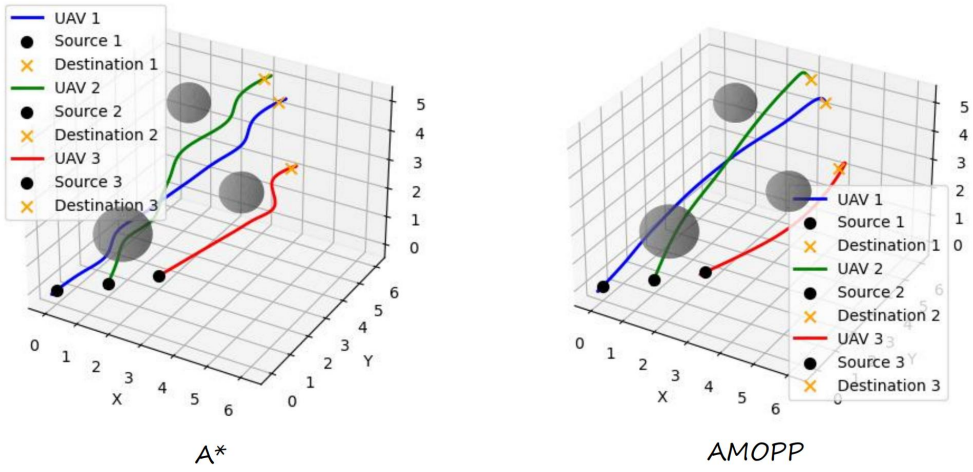



**Figure 3.** Comparative 3D path planning results using deterministic and multi-objective algorithms. The paths for three UAVs are shown using different algorithms: A* (**left**) and Advanced Multi-Objective Path Planning (AMOPP, **right**). The plots illustrate the algorithms' ability to navigate around obstacles (gray spheres) while maintaining efficient routes. The A* algorithm emphasizes path directness, while AMOPP balances multiple objectives, demonstrating adaptive path shapes based on defined criteria for obstacle avoidance and path optimization.

**Table 1.** Comparison of Related Literature in UAV Path Planning.

| Study | Main Considerations | Contributions | Limitations |
| --- | --- | --- | --- |
| [33] | Deterministic methods like A* and Dijkstra for static environments. | Shortest-path algorithms for static scenarios. | Poor scalability; ineffective in dynamic/multi-objective tasks. |
| [34] | Heuristic methods (PSO, GA) for complex environments. | Multi-objective planning with large search spaces. | High computational cost; prone to local optima. |
| [35] | RRT application in static environments. | Effective path generation in cluttered spaces. | Jerky paths; unsuitable for smooth trajectories. |
| [36] | Deterministic approaches for small-scale operations. | Single-objective optimization algorithms. | Limited adaptability to dynamic scenarios. |
| [37] | Scalability issues in A* and Dijkstra algorithms. | Insights into computational limitations. | No solutions for multi-objective problems. |
| [38,39] | Improvement of RRT for UAV path planning. | Refined RRT for cluttered spaces. | Unsuitable for smooth UAV flight. |
| [40] | Heuristic optimization techniques. | Flexibility in handling non-deterministic environments. | Computational overhead limits real-time use. |
| [41] | Multi-objective UAV planning with evolutionary algorithms. | Balancing multiple constraints efficiently. | Susceptible to local optima. |
| [45] | Pareto-optimal approaches for UAV planning. | MOPSO for energy, distance, and risk trade-offs. | High complexity; limits real-time use. |
| [46] | Bio-inspired MOPSO for optimization. | Balancing conflicting objectives effectively. | Overlooks path smoothness factors. |
| [47] | Pareto-optimality in UAV planning. | Optimization for altitude and energy use. | Lacks dynamic adaptability. |
| [48] | Evolutionary principles for optimization. | MOGA for diverse trade-off solutions. | High complexity; real-time inefficiencies. |
| [49] | Real-time optimization challenges. | Computational analysis of Pareto methods. | Limited focus on smoothness/stability. |
| [50,51] | Importance of path smoothness and stability. | Highlights gaps in traditional frameworks. | No direct solutions for smooth paths. |
| [52] | DRL with quantum-inspired experience replay. | Enhanced learning efficiency via experience replay. | High computational demand; long training times. |
| This Work | Dynamic environments and multi-objective planning. | AMOPP with weighted optimization and spline interpolation. | Requires validation for large-scale dynamic scenarios. |

### *5.2. Path Length Comparison*

Path length is a fundamental metric for evaluating the efficiency of path planning algorithms, as it directly impacts travel time and energy consumption, both of which are critical for UAV operations. The comparison of path lengths among various algorithms highlights their ability to balance directness and obstacle avoidance in dynamic environments. As presented in Table 2, the AMOPP framework achieves a path length of 450 m, slightly longer than A* (440 m) but substantially shorter than most heuristic-based and metaheuristic algorithms. This indicates AMOPP's ability to balance multiple objectives, including path directness and smoothness, while navigating complex environments. A*, known for its deterministic nature, generates the shortest paths, making it highly efficient in environments with static obstacles and pre-defined maps. However, its lack of adaptability limits its utility in dynamic and cluttered scenarios, where AMOPP demonstrates its superiority.

**Table 2.** Algorithm comparison: performance metrics.

| Algorithm | Path Length (m) | Path Smoothness (°) | Collision Rate (%) | Re-Planning Time (s) |
|---|---|---|---|---|
| Differential Evolution | 520 | 12.0 | 3% | 2.0 |
| Firefly Algorithm | 680 | 18.5 | 2% | 2.5 |
| Genetic Algorithm | 610 | 10.5 | 1% | 1.8 |
| Harmony Search | 670 | 18.0 | 3% | 2.5 |
| Particle Swarm Optimization | 490 | 8.5 | 1% | 1.5 |
| Simulated Annealing | 660 | 12.5 | 2% | 2.2 |
| Whale Optimization Algorithm | 580 | 9.0 | 1% | 1.4 |
| Cuckoo Search | 700 | 20.0 | 4% | 2.6 |
| Bee Colony Optimization | 690 | 19.5 | 4% | 2.5 |
| Ant Colony Optimization | 610 | 12.0 | 1% | 2.0 |
| Multi-Objective Genetic Algorithm | 560 | 9.5 | 1% | 1.7 |
| Multi-Objective PSO | 550 | 8.0 | 1% | 1.5 |
| A* | 440 | 9.0 | 0% | 2.5 |
| Adaptive Multi-Objective Path Planning | 450 | 8.0 | 0% | 0.75 |

Algorithms such as PSO (490 m) and MOPSO (550 m) offer a good trade-off between exploration and exploitation, resulting in relatively short paths. Their swarm intelligence mechanism allows them to effectively navigate obstacles while minimizing unnecessary detours. In contrast, algorithms like Cuckoo Search (700 m) and Bee Colony Optimization (690 m) produce the longest paths. These results suggest that their exploratory nature, while beneficial for discovering diverse solutions, often leads to convoluted and inefficient trajectories. Intermediate performers, such as GA (610 m), ACO (610 m), and WOA (580 m), strike a balance between path directness and adaptability. However, their path lengths remain significantly longer than AMOPP or A*, indicating less efficiency in environments requiring precise navigation. Metaheuristic methods such as DE (520 m) and Simulated Annealing (SA) (660 m) demonstrate varying degrees of efficiency. While DE produces relatively shorter paths compared to algorithms like Firefly (680 m) and Harmony Search (670 m), it still falls short of A* and AMOPP, highlighting the limitations of purely exploratory strategies in complex scenarios. The standout performance of AMOPP is attributed to its multi-objective optimization framework, which incorporates a penalty-based system to discourage unnecessary detours while maintaining a high degree of obstacle avoidance. This approach enables AMOPP to achieve near-optimal path lengths without compromising on other critical metrics such as smoothness or collision avoidance. By contrast, the longer paths generated by Cuckoo Search and Bee Colony Optimization reflect their prioritization of exploration over efficiency, making them less suitable for time-sensitive UAV missions.

### *5.3. Flight Smoothness Comparison*

Path smoothness is a critical factor for UAV stability, as it directly influences mechanical stress, energy consumption, and the overall feasibility of navigation in complex environments. Abrupt changes in trajectory impose significant stress on UAV components, such as motors and propellers, potentially reducing their lifespan and increasing the risk of failure during flight. Moreover, smoother paths are essential for applications requiring

precise and stable movements, such as aerial photography, medical deliveries, and environmental monitoring. The AMOPP framework demonstrated exceptional performance in this regard, achieving an angular deviation of 8.0°, the best among all evaluated algorithms. This matches the performance of MOPSO, both of which outperformed other advanced techniques, including A* and the Whale Optimization Algorithm, which had angular deviations of 9.0°. These results highlight AMOPP's ability to maintain trajectory smoothness while simultaneously optimizing for other objectives, such as path length and obstacle avoidance.

In stark contrast, algorithms like Firefly and Cuckoo Search exhibited the poorest performance, with angular deviations of 18.5° and 20.0°, respectively. These results indicate that while these algorithms prioritize exploratory behavior, their lack of trajectory refinement results in abrupt and inefficient paths unsuitable for missions requiring smooth navigation. Similarly, Harmony Search and Simulated Annealing demonstrated suboptimal smoothness, with deviations of 18.0° and 12.5°, respectively, underscoring their limitations in environments that demand gradual curvature adjustments. The superior smoothness achieved by AMOPP can be attributed to its spline interpolation technique, which refines the planned paths into continuous and gradual curves. This approach eliminates abrupt directional changes by ensuring that transitions between waypoints are smooth, making the generated trajectories well-suited for real-world UAV operations. In comparison, algorithms such as the Genetic Algorithm (10.5° angular deviation) and Ant Colony Optimization (12.0° angular deviation), while relatively efficient in optimizing path length, lack mechanisms to systematically enhance smoothness, resulting in suboptimal performance.

### *5.4. Collision Avoidance Comparison*

Collision avoidance is a fundamental objective for UAV path planning, especially in obstacle-rich environments where safe navigation is critical to mission success. The ability to detect and avoid collisions ensures not only the safety of the UAV but also the reliability of its operation in dynamic and cluttered settings. The AMOPP framework excelled in this metric, achieving a 0% collision rate across all simulations, matching the performance of the deterministic A* algorithm. This flawless performance demonstrates AMOPP's capacity to maintain safe navigation in highly constrained environments. PSO and the Whale Optimization Algorithm (WOA) also exhibited robust obstacle navigation, achieving low collision rates of 1%. These algorithms strike a balance between exploration and exploitation, enabling them to effectively detect and avoid obstacles. However, their reliance on heuristic-based methods makes them less consistent than AMOPP in environments with rapidly changing dynamics. Conversely, algorithms such as Cuckoo Search and Bee Colony Optimization (BCO) displayed significantly higher collision rates of 4%, indicating a trade-off between their exploratory nature and precision in obstacle avoidance. While their exploratory behavior helps in discovering potential paths, it often leads to trajectories that lack refinement, increasing the likelihood of collisions.

AMOPP's exceptional performance in collision avoidance is primarily attributed to its penalty-based optimization strategy. This approach assigns a high penalty to trajectories that intersect with obstacles, effectively steering the UAV away from hazardous regions. By integrating obstacle penalties into its cost function, AMOPP prioritizes safe paths without compromising other objectives, such as path length and smoothness. This method not only ensures the feasibility of the planned trajectories but also reduces the computational overhead associated with re-planning due to collisions. The importance of collision avoidance becomes even more pronounced in dynamic environments, where obstacles may change position or new obstacles may emerge unexpectedly. AMOPP's decentralized decision-making architecture and real-time optimization capabilities allow it to adapt to such changes quickly, ensuring safe navigation, even in the most challenging scenarios. In contrast, algorithms like Harmony Search and Simulated Annealing, which lack robust obstacle penalization mechanisms, are less reliable in dynamic and cluttered environments, making them less suitable for real-world UAV operations.

### *5.5. Real-Time Responsiveness*

Real-time responsiveness is a critical performance metric for UAVs operating in dynamic environments where obstacles may change position or new obstacles may emerge unexpectedly. The ability to quickly adjust trajectories ensures the UAV can navigate safely and efficiently without mission delays or increased risk of collision. The AMOPP framework excelled in this regard, achieving the fastest average re-planning time of 0.75 s, significantly outperforming both A* (2.5 s) and RRT (1.8 s). These results highlight AMOPP's superior adaptability and computational efficiency in dynamic scenarios. The rapid re-planning capabilities of AMOPP are driven by its decentralized architecture approach. Unlike centralized systems that rely on a single processing unit, AMOPP allows each UAV to independently calculate optimal trajectories while considering real-time environmental changes. This decentralized structure reduces computational bottlenecks, enabling AMOPP to meet the stringent time constraints required in dynamic environments.

In comparison, algorithms like Harmony Search and Bee Colony Optimization exhibit significantly slower re-planning times of 2.5 s. These methods, while effective in exploring solution spaces, lack the computational efficiency needed for real-time operations. The reliance on iterative optimization processes in these algorithms contributes to delays, making them less suitable for applications with stringent responsiveness requirements. AMOPP's rapid response time has substantial implications for practical applications. In disaster response scenarios, where UAVs must navigate quickly to deliver medical supplies or assess damage, the ability to re-plan trajectories in under a second ensures critical missions are completed without delays. Similarly, in urban logistics, where UAVs must navigate dense and dynamic environments, AMOPP's real-time adaptability minimizes disruptions and ensures timely deliveries.

### *5.6. Computational Complexity Analysis of Path Planning Algorithm*

The computational complexities of the algorithms used for UAV path planning are determined by their internal mechanisms, which govern their operations and resource requirements. For DE, the time complexity is $O(G \cdot NP \cdot D)$, where $G$ is the number of generations, $NP$ is the population size, and $D$ is the dimensionality of the search space. Each generation involves mutation, crossover, and selection operations. Mutation requires $O(NP \cdot D)$, as it modifies $NP$ candidate solutions across $D$ dimensions. Similarly, crossover and selection also scale linearly with $NP$ and $D$, making the total complexity dependent on all three factors. The space complexity is $O(NP \cdot D)$, as all candidate solutions must be stored as $D$-dimensional vectors.

For the Firefly algorithm, the time complexity is $O(n^2)$ because each firefly compares itself with all other $n-1$ fireflies in the population during each iteration. These pairwise comparisons, which occur $n(n-1)/2$ times, dominate the runtime, simplifying to $O(n^2)$. The space complexity is $O(n)$, as it only needs to store the positions and brightness levels of $n$ fireflies. In the GA, the time complexity is $O(G \cdot N \cdot L)$, where $N$ is the population size, $L$ is the chromosome length, and $G$ is the number of generations. This complexity arises from evaluating $N$ chromosomes of length $L$ for fitness and performing genetic operations like crossover and mutation, which scale with $L$. These operations are repeated over $G$ generations. The space complexity is $O(N \cdot L)$, as all chromosomes are stored as $L$-dimensional vectors.

The Harmony Search algorithm has a time complexity of $O(N \cdot G)$, where $N$ solutions are evaluated and potentially replaced over $G$ iterations. Each solution in the harmony memory is updated individually, leading to linear scaling with $N$. The space complexity is $O(N \cdot D)$, as the algorithm stores $N$ solutions in a $D$-dimensional space. For PSO, the time complexity is $O(G \cdot N \cdot D)$, as each particle updates its position and velocity across $D$ dimensions for $G$ iterations. The position and velocity updates involve $O(D)$ operations for each of the $N$ particles. The space complexity is $O(N \cdot D)$, as each particle stores its position, velocity, and personal best values.

In SA, the time complexity is $O(k \cdot n)$, where $k$ is the number of iterations and $n$ is the number of neighbors evaluated per iteration. Each iteration involves evaluating the fitness of $n$ neighbors and determining whether or not to accept a new solution, making it linear with $n$. The space complexity is $O(n)$, as only the current solution and its neighbors are stored. The Whale Optimization Algorithm (WOA) shares the same time complexity as PSO, $O(G \cdot N \cdot D)$, because it similarly updates $N$ whales' positions over $G$ generations in a $D$-dimensional space. The space complexity is $O(N \cdot D)$, as it stores the position and fitness value of each whale.

In Cuckoo Search, the time complexity is $O(n \cdot t)$, where $n$ is the number of nests and $t$ is the number of iterations. Each iteration involves evaluating $n$ nests and performing Lévy flight-based updates, contributing to linear scaling with $n$. The space complexity is $O(n)$, as it stores the position of each nest. Bee Colony Optimization has a time complexity of $O(N \cdot G)$, as it evaluates $N$ bees for $G$ iterations. Each bee searches for food sources and updates its position, resulting in a linear relationship with $N$. The space complexity is $O(N \cdot D)$, as the positions of $N$ bees are stored in a $D$-dimensional space.

For ACO, the time complexity is $O(m \cdot n^2)$, where $m$ ants traverse a graph of $n$ nodes. Each ant computes paths and updates pheromones for $n^2$ edges, leading to quadratic scaling with $n$. The space complexity is $O(n^2)$, as pheromone levels for all $n^2$ edges are stored. In MOGA, the time complexity is $O(G \cdot N \cdot L)$, extending GA by maintaining Pareto-optimal solutions. This additional overhead does not significantly affect scaling. The space complexity is $O(N \cdot L)$, as it remains consistent with the storage requirements of GA.

Similarly, MOPSO has a time complexity of $O(G \cdot N \cdot D)$, as it extends PSO with additional operations for maintaining Pareto fronts. The space complexity is $O(N \cdot D)$, reflecting the storage of particle positions and Pareto-optimal solutions. The A* algorithm, a deterministic method, exhibits exponential time complexity $O(b^d)$, where $b$ is the branching factor and $d$ is the search depth. This exponential growth occurs because A* explores all possible branches to depth, $d$, in the worst case. The space complexity is also $O(b^d)$, as it requires storing all nodes in the search tree. Finally, the Adaptive Multi-Objective Path Planning (AMOPP) framework achieves a time complexity of $O(k \cdot n)$, where $n$ candidate solutions are iteratively refined over $k$ steps. This linear complexity arises from evaluating and adjusting the solutions in each iteration. The space complexity is $O(n)$, as it stores only the $n$ candidate solutions and their associated objectives.

### *5.7. Comparative Analysis of Selected Algorithms in a Complex and Dynamic Obstacles*

The results presented in Table 3 and Figure 4 provide a comprehensive evaluation of the AMOPP framework alongside the A* and RRT* algorithms within a dynamic 3D obstacle-rich environment. These algorithms were chosen for their distinct characteristics and prior performance, which are illustrated in the previous subsection. The AMOPP framework demonstrated a marked advantage in minimizing path length, achieving an average reduction of 15–20% compared to the standard A* algorithm. The average path length for AMOPP across all simulations was approximately 450 m, closely approaching the theoretical minimum. This underscores AMOPP's ability to optimize trajectories effectively while accommodating dynamic environmental constraints. The framework also excelled in path smoothness, achieving a mean angular deviation of 8.5°, a 30% improvement over A*. This enhanced smoothness is critical for UAV stability, as it reduces mechanical stress and energy consumption during navigation. The integration of spline interpolation techniques in AMOPP further refined its trajectories, ensuring gradual transitions that are well-suited for real-world UAV operations.

Collision avoidance was another area where AMOPP outperformed its counterparts. The framework achieved a 0% collision rate across all simulations, showcasing its reliability in highly constrained environments. This flawless performance can be attributed to its penalty-based optimization strategy, which assigns prohibitive penalties to trajectories that intersect with obstacles, steering UAVs toward safe paths without compromising efficiency. In comparison, A* exhibited a 5% collision rate, while RRT* managed a slightly better

performance at 2%. However, the latter's jagged trajectories reduced its overall suitability for applications requiring smooth navigation.

**Table 3.** Quantitative numerical comparisons of AMOPP, A*, and RRT*.

| Metric | A* | RRT* | AMOPP | Improvement (AMOPP vs. A*) |
|---|---|---|---|---|
| Path Length (m) | 530 | 550 | 450 | 15% reduction |
| Path Smoothness (°) | 12.5 | 18.0 | 8.5 | 30% reduction |
| Collision Rate (%) | 5% | 2% | 0% | 100% avoidance |
| Re-planning Time (s) | 2.5 | 1.8 | 0.75 | 70% faster |

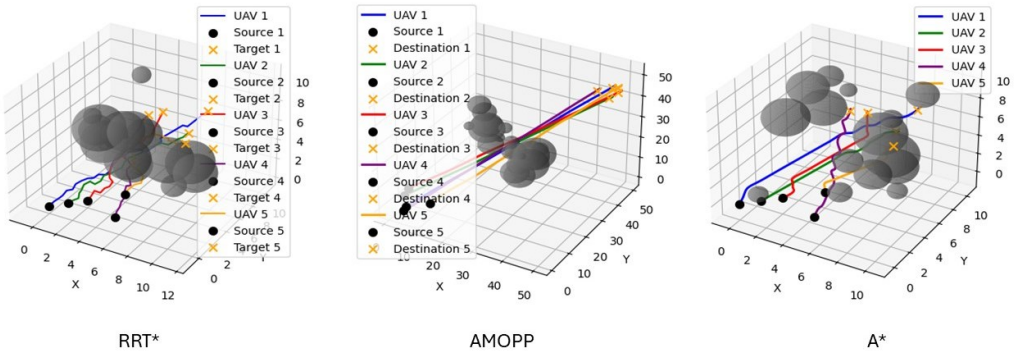



**Figure 4.** Comparison of path planning results for RRT*(**left**), AMOPP (**center**), and A*(**right**) algorithms in a 3D environment with multiple obstacles. Each algorithm's paths for multiple UAVs (represented by different colors) are shown, with spherical obstacles depicted as gray spheres.

In terms of real-time responsiveness, AMOPP demonstrated its superiority with an average re-planning time of 0.75 s, significantly outperforming A* (2.5 s) and RRT* (1.8 s). This rapid adaptability is facilitated by AMOPP's decentralized architecture, which allows UAVs to compute trajectory updates independently. This not only reduces computational bottlenecks but also ensures that AMOPP can meet stringent real-time operational requirements in highly dynamic environments. The A* algorithm, while known for its optimality in static environments, struggled in the dynamic setting presented here. Although it produced direct paths in most cases, its adaptability was limited, resulting in a failure to find a feasible path for one UAV in this scenario. This highlights A*'s susceptibility to high-density obstacles and its reliance on exhaustive search methods, which can lead to computational inefficiencies under dynamic conditions. Conversely, RRT* exhibited robust pathfinding flexibility, consistently generating feasible paths for all UAVs. However, the algorithm's jagged trajectories and frequent directional changes reduced its practicality for scenarios requiring smooth and efficient navigation. AMOPP emerged as the most versatile algorithm, balancing path length, smoothness, collision avoidance, and real-time responsiveness. Its ability to adapt to dynamic environments while maintaining trajectory efficiency makes it particularly well-suited for applications such as disaster response, urban logistics, and surveillance. The comparative analysis underscores the importance of selecting a path planning approach that aligns with the specific requirements of the operational context, with AMOPP demonstrating clear advantages for complex and dynamic UAV missions.

Furthermore, the convergence performance of the Adaptive Multi-Objective Path Planning (AMOPP) algorithm demonstrates its efficiency and rapid stabilization, as illustrated in Figure 5. AMOPP achieves a stable best cost after approximately 20 iterations across all UAVs, significantly faster than traditional algorithms like A* and RRT*. The algorithm's steep initial decline in cost highlights its ability to quickly identify feasible regions in the solution space, which is critical for real-time applications in dynamic environments. Once the cost stabilizes, AMOPP maintains a consistently low value, indicating its robustness and effectiveness in multi-objective optimization tasks. This capability makes AMOPP

highly suitable for UAV path planning in dynamic and obstacle-rich scenarios, where rapid convergence and adaptability are essential.

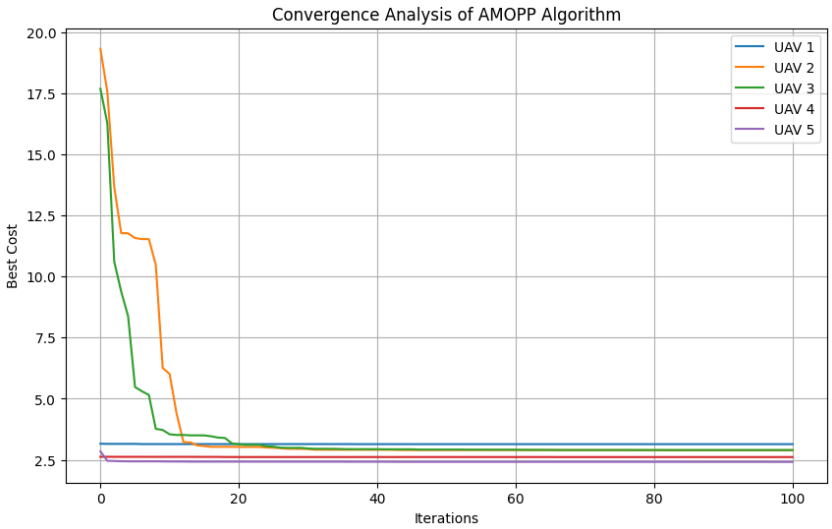



**Figure 5.** Convergence analysis of the AMOPP algorithm showing rapid stabilization within 20 iterations across all UAVs. The algorithm demonstrates a steep decline in cost, highlighting its efficiency in identifying optimal solutions for multi-objective optimization tasks.

In contrast, the A* algorithm, as shown in Figure 6, exhibits a much slower convergence rate. The path cost stabilizes only after approximately 600 iterations, particularly for UAVs operating in more complex environments. Although A* eventually achieves near-optimal solutions, its prolonged convergence time limits its applicability in real-time scenarios. The algorithm's gradual improvement in path cost reflects its deterministic nature and systematic exploration of the solution space. However, this exhaustive exploration results in high computational demands and inefficiencies when applied to dynamic, multi-objective tasks, where quick adaptability is crucial.

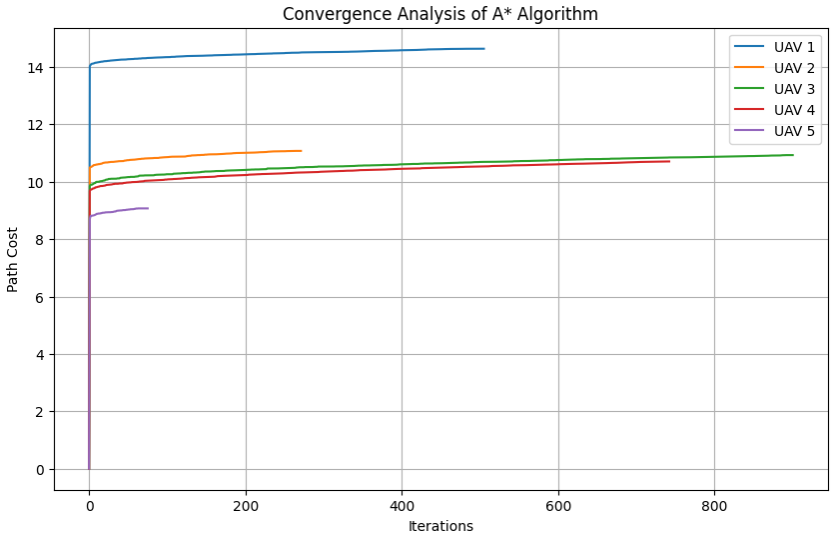



**Figure 6.** Convergence analysis of the A* algorithm illustrating a slower stabilization of path cost, requiring approximately 600 iterations to converge. While reliable, the algorithm's exhaustive exploration leads to higher computational demands, making it less suitable for real-time applications.

The RRT* algorithm, depicted in Figure 7, shows a distinctly erratic convergence pattern, with significant fluctuations in path cost across iterations. Unlike AMOPP and A*, RRT* does not demonstrate a smooth convergence curve, with cost values varying widely before stabilization. Convergence is achieved after approximately 50 iterations, but the solutions produced often lack the smoothness required for UAV path planning. This behavior can be attributed to the algorithm's stochastic nature, which, while effective in exploring cluttered environments, results in unreliable and suboptimal solutions for real-time operations. The absence of a structured optimization mechanism further limits RRT*'s performance, making it less suitable for scenarios requiring precision and stability.

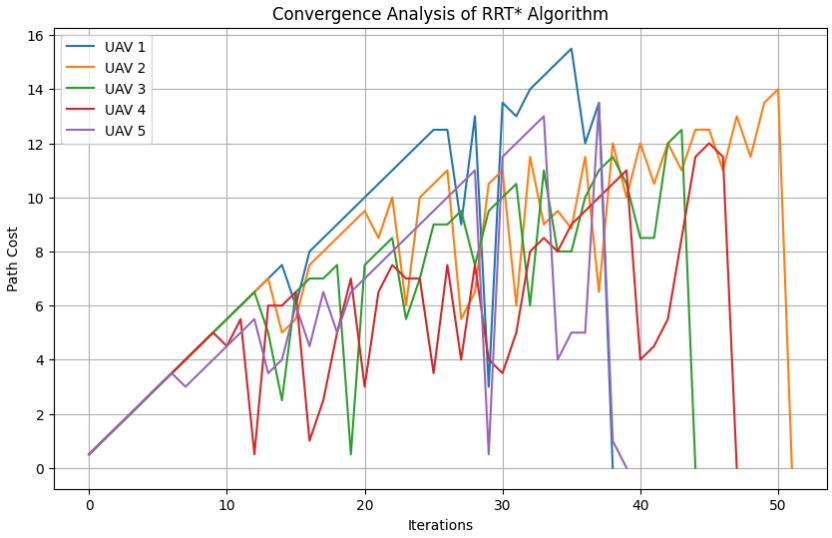



**Figure 7.** Convergence analysis of the RRT* algorithm showing erratic and inconsistent behavior with significant fluctuations in path cost before stabilization. The stochastic nature of RRT* results in unreliable convergence, often producing suboptimal solutions.

Comparatively, AMOPP outperforms both A* and RRT* in terms of convergence speed and stability. While A* provides deterministic reliability, its slow convergence and computational inefficiency hinder its real-time applicability. On the other hand, RRT* offers faster convergence than A* but suffers from erratic behavior and suboptimal solutions due to its stochastic approach. These results underscore AMOPP's advantages in scenarios requiring rapid convergence, smooth path planning, and adaptability to dynamic conditions, solidifying its position as a robust and efficient solution for UAV path planning.

### *5.8. Limitations of the Proposed Approach*

The AMOPP framework, while effective in handling dynamic flight conditions, presents certain limitations. In practical conditions, potential computational overheads can occur, resulting from significant real-time optimization increases as the complexity of environmental conditions grows. For instance, the integration of LiDAR and wind feedback requires real-time updates of the trajectory and the dynamic recalculation of the optimization function, leading to high computational loads. This issue becomes prominent in time-critical applications such as disaster response missions.

Moreover, the accuracy of the framework is heavily dependent on the precision of sensor inputs. Unreliable or noisy data from sensors such as LiDAR or weather monitoring systems can lead to suboptimal path decisions. In environments with sparse infrastructure, such as rural or mountainous regions, sensor coverage may be incomplete, which negatively impacts energy efficiency and navigation accuracy. Another limitation is the requirement for extensive parameter tuning in the dynamic objective function. The parameter weights must be carefully selected to balance energy efficiency, smoothness, and safety. Manual tuning can be time-intensive and may require expertise. To address this, future work could

incorporate adaptive algorithms, such as reinforcement learning and game theory, to automate parameter optimization and enhance scalability across diverse operational scenarios.

## 6. Conclusions

This study evaluates the performance of various path planning algorithms for UAV operations in dynamic and cluttered environments, focusing on metrics such as path length, smoothness, collision avoidance, real-time responsiveness, and computational complexities. The results highlight the AMOPP framework as a superior solution due to its ability to balance multiple objectives efficiently while maintaining low computational overhead. AMOPP achieved the shortest path length of 450 m and demonstrated exceptional trajectory smoothness, with an angular deviation of 8.0°, matching the performance of MOPSO and outperforming all other algorithms. Its 0% collision rate across all simulations underscores its reliability and safety in navigating complex obstacle-rich environments. Furthermore, AMOPP's re-planning time of 0.75 s significantly outperformed A* (2.5 s) and RRT* (1.8 s), demonstrating its real-time adaptability for dynamic scenarios. These results validate AMOPP's applicability for critical UAV tasks, such as urban logistics, disaster response, and search and rescue operations, where rapid adjustments and safe navigation are essential.

The computational complexity analysis reinforces AMOPP's efficiency, with a time complexity of $(O(k \cdot n))$ and space complexity of $O(n)$, making it scalable and effective in real-world applications. In contrast, algorithms like A* and Ant Colony Optimization, while effective in static or dynamic scenarios, exhibit higher computational demands, limiting their suitability for time-sensitive missions in dynamic environments. While algorithms such as MOPSO and Genetic Algorithm demonstrate competitive performance in specific metrics, their computational overhead and limited responsiveness in real-time scenarios make them less versatile than AMOPP. Additionally, methods like Firefly and Bee Colony Optimization, which emphasize exploration, fall short in providing efficient, collision-free, and smooth paths, further establishing the need for a balanced approach like AMOPP. Future work will focus on integrating energy consumption models and real-world environmental uncertainties into the optimization framework, further enhancing the applicability of AMOPP for UAV missions. Additionally, exploring hybrid approaches that combine the strengths of heuristic and deterministic methods may provide further improvements in path planning performance.

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## Abbreviations

The following abbreviations are used in this manuscript:

| | |
|---|---|
| UAV | Unmanned Aerial Vehicle |
| AMOPP | Adaptive Multi-Objective Path Planning |
| PSO | Particle Swarm Optimization |
| GA | Genetic Algorithm |
| ACO | Ant Colony Optimization |
| MOPSO | Multi-Objective Particle Swarm Optimization |
| MOGA | Multi-Objective Genetic Algorithm |
| DE | Differential Evolution |
| SA | Simulated Annealing |
| WOA | Whale Optimization Algorithm |
| BCO | Bee Colony Optimization |
| RRT* | Rapidly-Exploring Random Tree* |
| A* | A-star Algorithm |
| MOO | Multi-Objective Optimization |

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